Supplementary Material for *Biological positive controls for cross-cohort validation of prognostic gene expression, with a case study of EGFR in glioma*

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This document is the supplementary material for the manuscript above. Every table and figure here is cited in the main text; numbering (S1, S2, . . .) is continuous with those citations.

Supplementary Methods: Computational Environment

All analyses were performed in R version 4.6.0 (2026-04-24) on macOS Tahoe 26.5.1 (Darwin 25.5.1, aarch64-apple-darwin24.6.0). Key package versions: **TCGAbiolinks** 2.40.0 (Bioconductor), **SummarizedExperiment** 1.42.0 (Bioconductor), **DESeq2** 1.52.0 (Bioconductor; VST), **survival** 3.8-6 (CRAN; Cox models, Kaplan-Meier), **survminer** 0.5.2 (CRAN; survival plotting), **dplyr** 1.1.4, and **ggplot2** 3.5.1 (CRAN). Matrix operations used OpenBLAS 0.3.33 (BLAS) and R's internal LAPACK 3.12.1. Full session information is archived with the analysis scripts at https://github.com/edanmn/TCGA-EGFR-Glioma.

Supplementary Figures

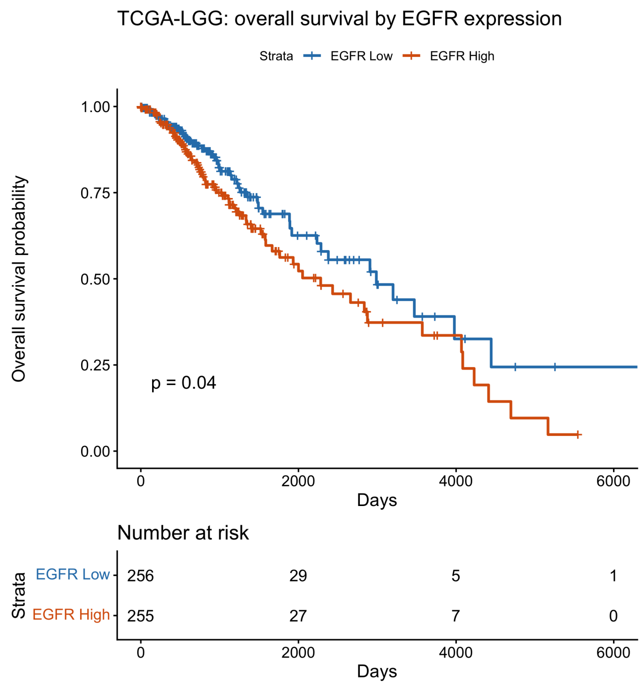


Figure S2. Unadjusted KM overall survival by EGFR group (median split), TCGA-LGG.

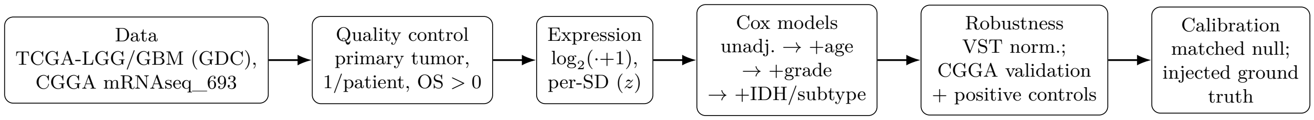
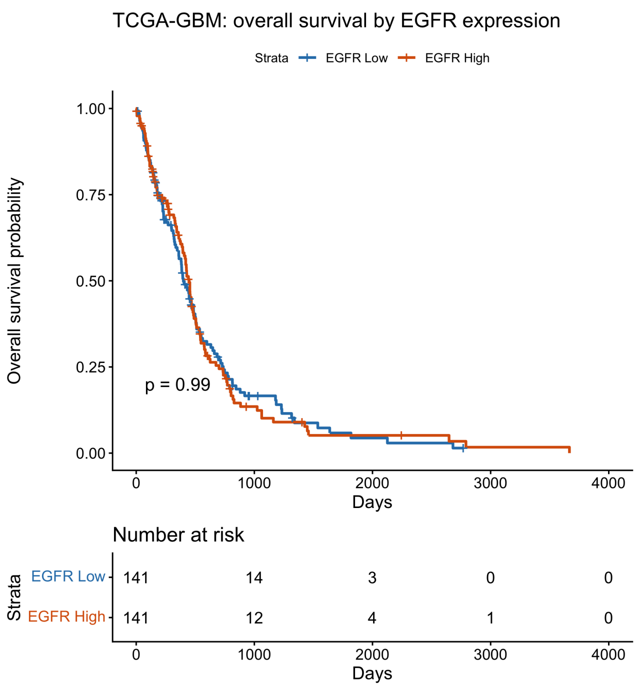


Figure S1. Analysis pipeline. Standardized expression enters nested Cox models with progressive confounder adjustment, followed by VST-normalization sensitivity and a positive-control–gated CGGA external-validation attempt, and finally the calibration stage added in revision: a matched same-population null and a controlled experiment with injected corruption at known doses (§6.9). Acquisition is cached separately from analysis.



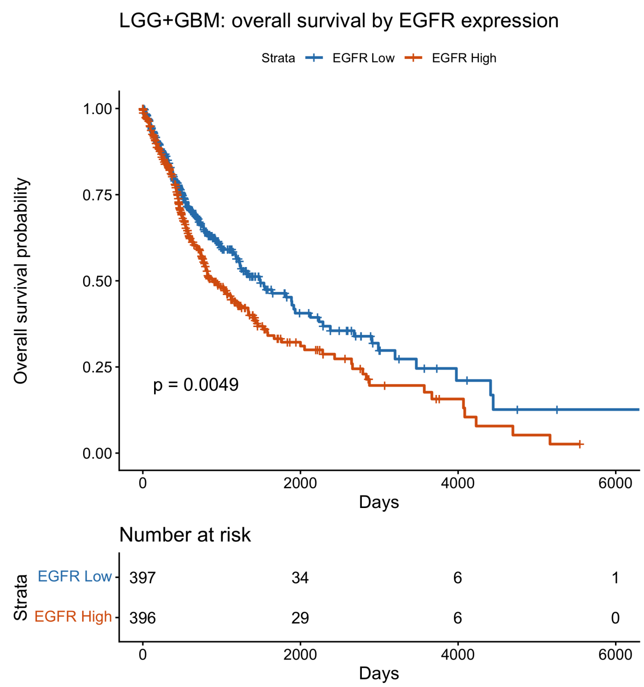


Figure S3. Unadjusted KM overall survival by EGFR group, TCGA-GBM.

Figure S4. Unadjusted KM overall survival by EGFR group, pooled LGG+GBM.

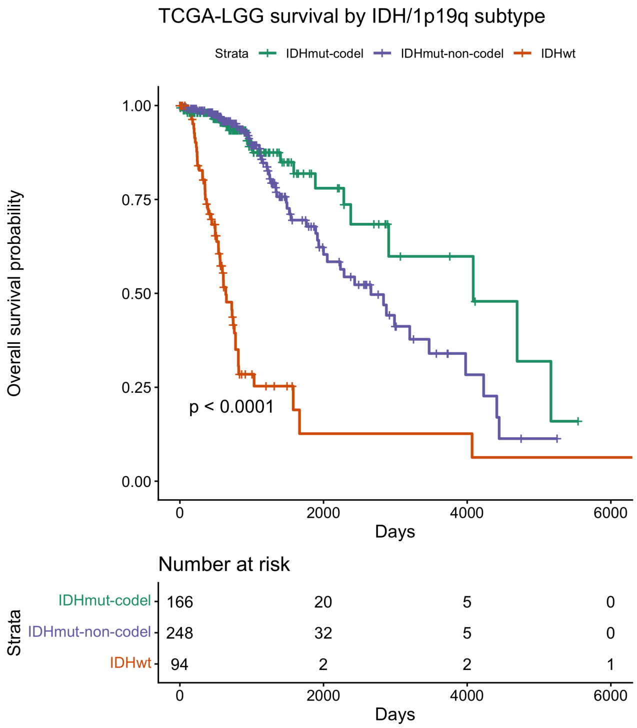


Figure S5. TCGA-LGG overall survival by IDH/1p19q molecular subtype; subtype, not EGFR, drives the separation in Figure S2.

Supplementary Tables

Supplementary Tables (moved from the main text in revision)

Supplementary Figures (moved from the main text in revision)

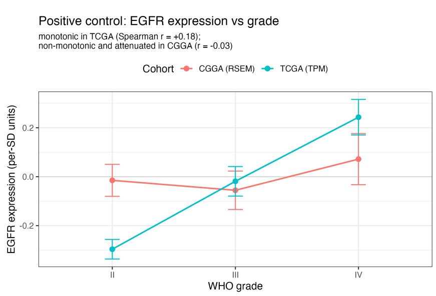


Figure S6. Positive control: standardized EGFR expression by WHO grade. EGFR rises monotonically with grade in TCGA (Spearman $r = +0.18$; canonical biology). In CGGA the relationship is *non-monotonic* rather than level—it falls from grade II to III ($-0.015 \rightarrow -0.055$) and rises from III to IV ($\rightarrow +0.072$), with overlapping standard errors—so the near-zero rank correlation ($r = -0.03$) reflects the absence of a consistent gradient, not a flat line. Points are means, error bars $\pm$ standard error. The between-cohort difference is significant (Fisher r-to-z $p = 4.5 \times 10^{-4}$). TCGA $n = 739$ (grade II: 216, III: 241, IV: 282); CGGA-693 $n = 422$ (II: 138, III: 144, IV: 140).

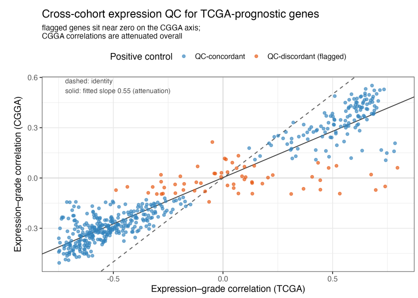


Figure S7. Cross-cohort expression QC for TCGA-prognostic genes. Each point is a gene; axes are its expression–grade correlation in TCGA (x) and CGGA (y). The dashed line is the identity; the solid line is the fitted slope of 0.55, showing that CGGA correlations are attenuated relative to TCGA's across the whole panel. Flagged genes (orange), including EGFR, are those sitting near zero on the CGGA axis, and are where cross-cohort prognostic replication breaks down. The flag marks discordance, not a demonstrated cause (§6.4).

Table S1. Unadjusted Kaplan–Meier median-split results for EGFR in TCGA (descriptive).

Cohort	n low / high	Events	Median survival, low vs. high (days)	Log-rank p
TCGA-LGG	256 / 255	125	2,988 vs. 2,282	0.040
TCGA-GBM	141 / 141	227	399 vs. 448	0.99
Pooled (LGG+GBM)	397 / 396	352	1,491 vs. 883	4.85×10^{-3}

Table S2. FDR-controlled pathway screen in TCGA-LGG (HR per 1 SD, age-adjusted), with FDR-adjusted p before and after adding *IDH* status.

Gene	HR (95% CI), age-adjusted	p_{BH} (age)	p_{BH} (age + IDH)
RB1	1.64 (1.34–2.00)	1.59×10^{-5}	0.031
IDH1	1.46 (1.20–1.78)	8.67×10^{-4}	0.033
EGFR	1.37 (1.14–1.66)	3.54×10^{-3}	0.178
TP53	1.39 (1.13–1.71)	5.20×10^{-3}	0.026
PIK3R1	0.777 (0.649–0.931)	1.25×10^{-2}	0.876
PTEN	0.800 (0.666–0.961)	2.83×10^{-2}	0.506
CDKN2A	0.834 (0.691–1.01)	8.72×10^{-2}	0.046
PIK3CA	1.13 (0.950–1.34)	0.212	0.046
PDGFRA	0.893 (0.752–1.06)	0.220	0.594
NF1	0.895 (0.750–1.07)	0.222	0.085

Table S3. Normalization sensitivity: EGFR HR per 1 SD under DESeq2 VST (compare to the TPM results in Table 1).

Cohort	Model	HR (95% CI)	p	C-index
TCGA-LGG	Unadjusted	1.50 (1.22–1.84)	1.25×10^{-4}	0.577
TCGA-LGG	+ age, grade	1.26 (1.04–1.54)	0.020	0.784
TCGA-LGG	+ age, grade, IDH	1.11 (0.944–1.31)	0.205	0.829
TCGA-GBM	Unadjusted	0.945 (0.829–1.08)	0.397	0.524
TCGA-GBM	+ age, IDH	0.843 (0.737–0.963)	0.012	0.641

Table S4. Positive controls: Spearman correlation of gene expression with WHO grade in TCGA vs. CGGA. EGFR (and the panel generally) tracks grade in TCGA but not CGGA, indicating attenuated expression–phenotype signal in the CGGA matrix. Correlations are computed on the same sample sets as Table 2, so the EGFR row matches that table exactly (+0.179 in TCGA, $n = 739$; −0.033 in CGGA-693, $n = 422$).

Gene	r(expression, grade), TCGA	r(expression, grade), CGGA
EGFR	+0.179	−0.033
IDH1	+0.348	+0.152
TP53	+0.099	+0.113
CDKN2A	−0.131	−0.076
NF1	−0.285	−0.183
PTEN	−0.357	−0.071
PDGFRA	−0.304	−0.096
PIK3R1	−0.481	−0.195
PIK3CA	−0.072	−0.057
RB1	−0.057	−0.023

Table S5. Pipeline robustness: nested EGFR Cox models (HR per 1 SD) on TCGA re-obtained from recount3 (independent Monorail raw-read reprocessing). Estimates match the GDC-pipeline results (Table 1).

Cohort	Model	HR (95% CI)	p	n / events
TCGA-LGG	Unadjusted	1.54 (1.25–1.89)	4.02×10^{-5}	509 / 125
TCGA-LGG	+ age, grade	1.27 (1.04–1.55)	0.021	452 / 106
TCGA-LGG	+ age, grade, IDH	1.11 (0.94–1.31)	0.220	450 / 105
TCGA-GBM	Unadjusted	0.906 (0.755–1.09)	0.289	154 / 122
TCGA-GBM	+ age	0.853 (0.712–1.02)	0.084	154 / 122
TCGA-GBM	+ age, grade, IDH	0.813 (0.677–0.977)	0.027	150 / 120

Table S6. Positive-control-gated cross-cohort replication for genes prognostic in TCGA (FDR < 0.05), among 500 top-variable genes, in two cohort pairs that share no patients. Replication is far higher for QC-concordant (grade-concordant) than QC-discordant (grade-discordant) genes; Fisher exact $p < 10^{-20}$ for both pairs. Both sign-flips observed in the panel fall among QC-discordant genes, though with only two such events this is suggestive rather than decisive.

Cohort pair	Positive control	Genes	Replicated % (95% CI)
TCGA → CGGA-693	QC-concordant	429	92.8 (89.9–95.0)
TCGA → CGGA-693	QC-discordant	37	24.3 (11.8–41.2)
TCGA → CGGA-325	QC-concordant	439	95.4 (93.1–97.2)
TCGA → CGGA-325	QC-discordant	26	19.2 (6.6–39.4)

Table S7. Circularity control (TCGA → CGGA-693, grade-anchor QC; bootstrap 95% CIs). The QC's predictive value is partly inflated by grade-correlated genes but persists for subtype-independent associations (grade+IDH-adjusted) and low-grade-correlation genes, excluding a purely tautological explanation. This control uses the TCGA∩CGGA-693 gene universe (6,616 prognostic genes) rather than the four-cohort intersection used for the headline analysis (6,512 genes; Table 4), which is why n, the replication rate, and the AUC differ slightly from Table 4.

Prognostic-gene set	Genes	Replication rate	AUC (95% CI)
Unadjusted (age only)	6,616	0.63	0.82 (0.81–0.83)
Subtype-adjusted (grade + IDH)	183	0.23	0.65 (0.55–0.74)
Low grade-correlation ($\|r\| < 0.2$)	935	0.51	0.75 (0.72–0.78)
High grade-correlation ($\|r\| \geq 0.2$)	5,681	0.65	0.84 (0.83–0.85)

Table S8. Naive CGGA external-validation models: EGFR HR per 1 SD in primary tumors. Retained for transparency only; these estimates are *not* interpretable, because CGGA EGFR expression fails the positive control (Table 2, Table S4, Figure 1). The apparent IDH-adjusted and treatment-adjusted effects are artifacts of EGFR mis-measurement in CGGA, not evidence of a prognostic effect.

Cohort	Model	HR (95% CI)	p	C-index
CGGA II/III	Unadjusted	1.17 (0.938–1.46)	0.164	0.520
CGGA II/III	+ age, grade	1.19 (0.958–1.48)	0.115	0.624
CGGA II/III	+ age, grade, IDH	1.44 (1.17–1.76)	4.6×10^{-4}	0.737
CGGA II/III	+ treatment (sens.)	1.37 (1.09–1.71)	6.8×10^{-3}	0.743
CGGA IV	+ age, IDH	1.12 (0.919–1.35)	0.268	0.594

Table S9. TCGA cohort characteristics and covariate availability after quality control. *IDH* status is available for 508 (LGG) and 265 (GBM) patients and molecular subtype for 508 and 257; counts sum to the cohort total including the unknown category. The grade row counts patients with a *recorded* histologic grade; all TCGA-GBM patients are treated as WHO grade IV by study definition, and one lacks a recorded grade field. Sex was not populated in the expression-object annotation.

Characteristic	TCGA-LGG	TCGA-GBM
Histologic grade	II–III	IV
Patients	511	282
Deaths (events)	125	227
Median age, years (IQR)	41 (32–53)	60 (51–69)
IDH (Mutant / WT / unknown)	414 / 94 / 3	21 / 244 / 17
Subtype (codel / non-codel / wt)	166 / 248 / 94	— / 19 / 238
Histologic grade recorded	454	281

Table S10. Nested Cox models for EGFR expression in TCGA (HR per 1 SD), with Harrell's C-index and the PH test p-value for the EGFR term. Grade is omitted for glioblastoma.

Cohort	Model	HR (95% CI)	p	C-index	PH p (EGFR)	n / events
TCGA-LGG	Unadjusted	1.59 (1.30–1.94)	6.85×10^{-6}	0.592	0.98	511 / 125
TCGA-LGG	+ age	1.37 (1.14–1.66)	1.06×10^{-3}	0.744	0.88	511 / 125
TCGA-LGG	+ age, grade	1.31 (1.07–1.59)	8.22×10^{-3}	0.786	0.23	454 / 106
TCGA-LGG	+ age, grade, IDH	1.13 (0.95–1.33)	0.164	0.830	0.004	452 / 105
TCGA-LGG	+ age, subtype	1.14 (0.98–1.34)	0.096	0.820	0.012	508 / 123
TCGA-GBM	Unadjusted	0.962 (0.845–1.09)	0.552	0.523	0.19	282 / 227
TCGA-GBM	+ age, IDH	0.866 (0.761–0.986)	0.030	0.639	0.40	265 / 214
Pooled	Unadjusted	1.35 (1.20–1.51)	1.98×10^{-7}	0.553	0.03	793 / 352
Pooled	+ age, grade, IDH	0.921 (0.840–1.01)	0.078	0.837	0.56	717 / 319

Table S11. EGFR positive control across TCGA and three CGGA datasets spanning two platforms, all computed by one method: Spearman correlation of EGFR expression with WHO grade among primary tumors with a recorded grade; Fisher r-to-z against the TCGA reference; univariable IDH-mutant HR; and grade IV vs. grade II HR (grade II and IV patients only). The correlation is positive in TCGA but null/negative in every CGGA dataset, whereas clinical controls are strong and correct throughout. The expression-level failure is reproducible across batches and platforms; the clinical data are sound.

Cohort	r(EGFR, grade)	n	p vs. TCGA	IDH-mutant HR	Grade IV vs. II HR
TCGA (reference)	+0.179	739	—	0.11	14.8
CGGA-693 (RNA-seq)	−0.033	422	4.5×10^{-4}	0.23	8.0
CGGA-325 (RNA-seq)	−0.097	229	2.5×10^{-4}	0.22	9.3
CGGA-301 (microarray)	−0.086	264	2.0×10^{-4}	0.32	7.3

Table S12. EGFR is not a within-subtype prognostic factor in IDH-wildtype glioma (TCGA). Expression (per 1 SD) shows a well-powered null; copy-number amplification shows no evidence of an effect (continuous, reasonably powered) and null across binary thresholds, though binary tests are underpowered (80% power only for HR ≥ 1.65). Grade is the pre-2021 histologic grade.

Marker	Model	HR (95% CI)	p	n / events
Expression	Unadjusted	1.00 (0.89–1.13)	0.99	338 / 253
Expression	+ age	0.93 (0.82–1.05)	0.22	338 / 253
Expression	+ age, grade	0.90 (0.80–1.02)	0.10	330 / 246
Expression	IDH-wt GBM, + age	0.87 (0.76–0.99)	0.034	244 / 203
Amplification	$\log_2$ copy number (per SD), +age	1.00 (0.84–1.20)	0.98	166 / 127
Amplification	CN ≥ 5 vs. not, +age	1.09 (0.76–1.57)	0.63	166 / 127
Amplification	CN ≥ 6 vs. not, +age	1.00 (0.70–1.42)	0.98	166 / 127
Amplification	CN ≥ 7 vs. not, +age	0.92 (0.65–1.32)	0.66	166 / 127